Problem | Editorial | Submissions | Comments

Subset Sums

Solved

Difficulty: **Medium** Accuracy: **72.55%** Submissions: **197K+** Points: **4**

Given an array **arr** of integers, return the sums of all subsets in the list. Return the sums in any order.

Examples:

Input: arr[] = [2, 3]
Output: [0, 2, 3, 5]
Explanation: When no elements are taken then Sum = 0. When only 2 is taken then Sum = 2. When only 3 is taken then Sum = 3. When elements 2 and 3 are taken then Sum = 2+3 = 5.

Input: arr[] = [1, 2, 1]
Output: [0, 1, 1, 2, 2, 3, 3, 4]
Explanation: The possible subset sums are 0 (no elements), 1 (either of the 1's), 2 (the element 2), and their combinations.

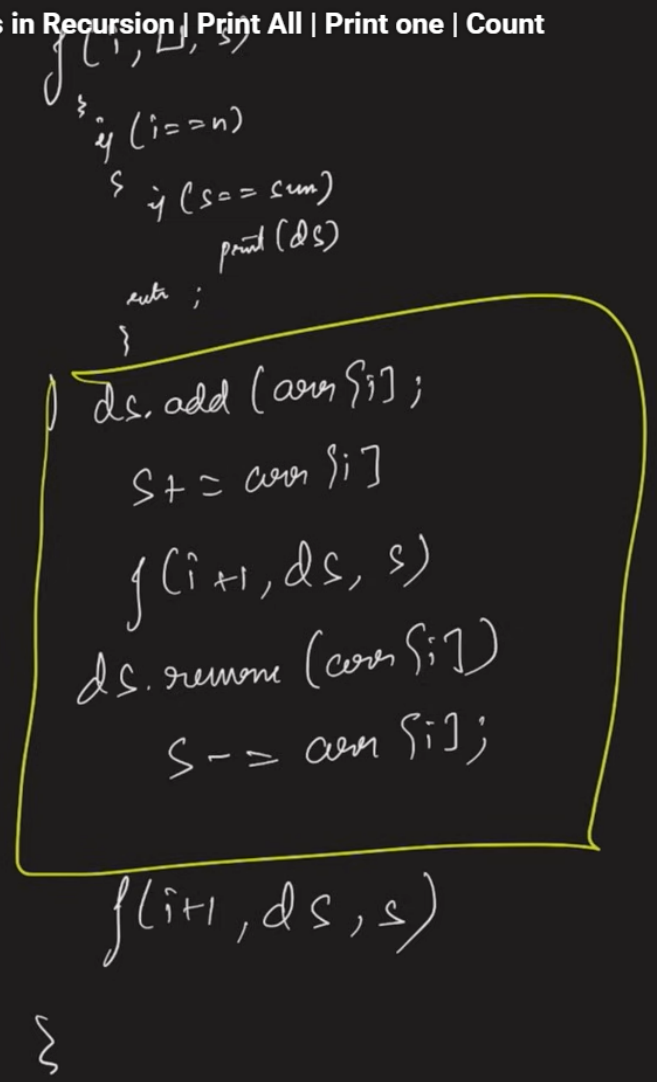
Input: arr[] = [5, 6, 7]
Output: [0, 5, 6, 7, 11, 12, 13, 18]
Explanation: The possible subset sums are 0 (no elements), 5, 6, 7, and their combinations.

C++ (17) | Start Timer

```cpp
class Solution {
public:
    void subset(int ind, vector<int>& ds, vector<int>& arr, int n, vector<int>& ans) {
        if (ind == n) {
            int sum = 0;

            for (auto it : ds) {
                sum += it;
            }

            ans.push_back(sum);
            return;
        }

        ds.push_back(arr[ind]);
        subset(ind + 1, ds, arr, n, ans);

        ds.pop_back();
        subset(ind + 1, ds, arr, n, ans);
    }

    vector<int> subsetSums(vector<int>& arr) {
        vector<int> ds;
        vector<int> ans;
        int n = arr.size();

        subset(0, ds, arr, n, ans);

        return ans;
    }
};
```

Custom Input | Compile & Run | Submit

L7. All Kind of Patterns in Recursion | Print All | Print one | Count



L7. All Kind of Patterns in Recursion | Print All | Print one | Count

code.cpp

```cpp
        if(s == sum) {
            for(auto it : ds) cout << it << " ";
            cout << endl;
        }
        return;
    }

    ds.push_back(arr[ind]);
    s += arr[ind];

    printS(ind+1, ds, s, sum, arr, n);

    s-= arr[ind];
    ds.pop_back();

    // not pick
    printS(ind+1,ds, s, sum, arr, n);
}
int main() {
    #ifndef ONLINE_JUDGE
    freopen("input.txt", "r", stdin);
    freopen("output.txt", "w", stdout);
    #endif
    int arr[] = {1, 2, 1};
    int n = 3;
    int sum = 2;
    vector<int> ds;
    printS(0, ds, 0, sum, arr, n);

    return 0;
}
```

input.txt

```
5
1 2 3 4 5
```

output.txt

```
1 1
2
```

```
[Finished in 1.9s]
```

11:03 / 34:11 Code

L7. All Kind of Patterns in Recursion | Print All | Print one | Count

code.cpp

```cpp
#include<bits/stdc++.h>
using namespace std;

bool flag = false;
void printS(int ind, vector<int> &ds, int s, int sum, int arr[], int n) {
    if(ind == n) {
        if(s == sum && flag == false) {
            flag = true;
            for(auto it : ds) cout << it << " ";
            cout << endl;
        }
        return;
    }

    ds.push_back(arr[ind]);
    s += arr[ind];

    printS(ind+1, ds, s, sum, arr, n);

    s-= arr[ind];
    ds.pop_back();

    // not pick
    printS(ind+1,ds, s, sum, arr, n);
}
int main() {
    #ifndef ONLINE_JUDGE
    freopen("input.txt", "r", stdin);
    freopen("output.txt", "w", stdout);
    #endif
    int arr[] = {1, 2, 1};
    int n = 3;
    int sum = 2;
    vector<int> ds;
    printS(0, ds, 0, sum, arr, n);
```

input.txt

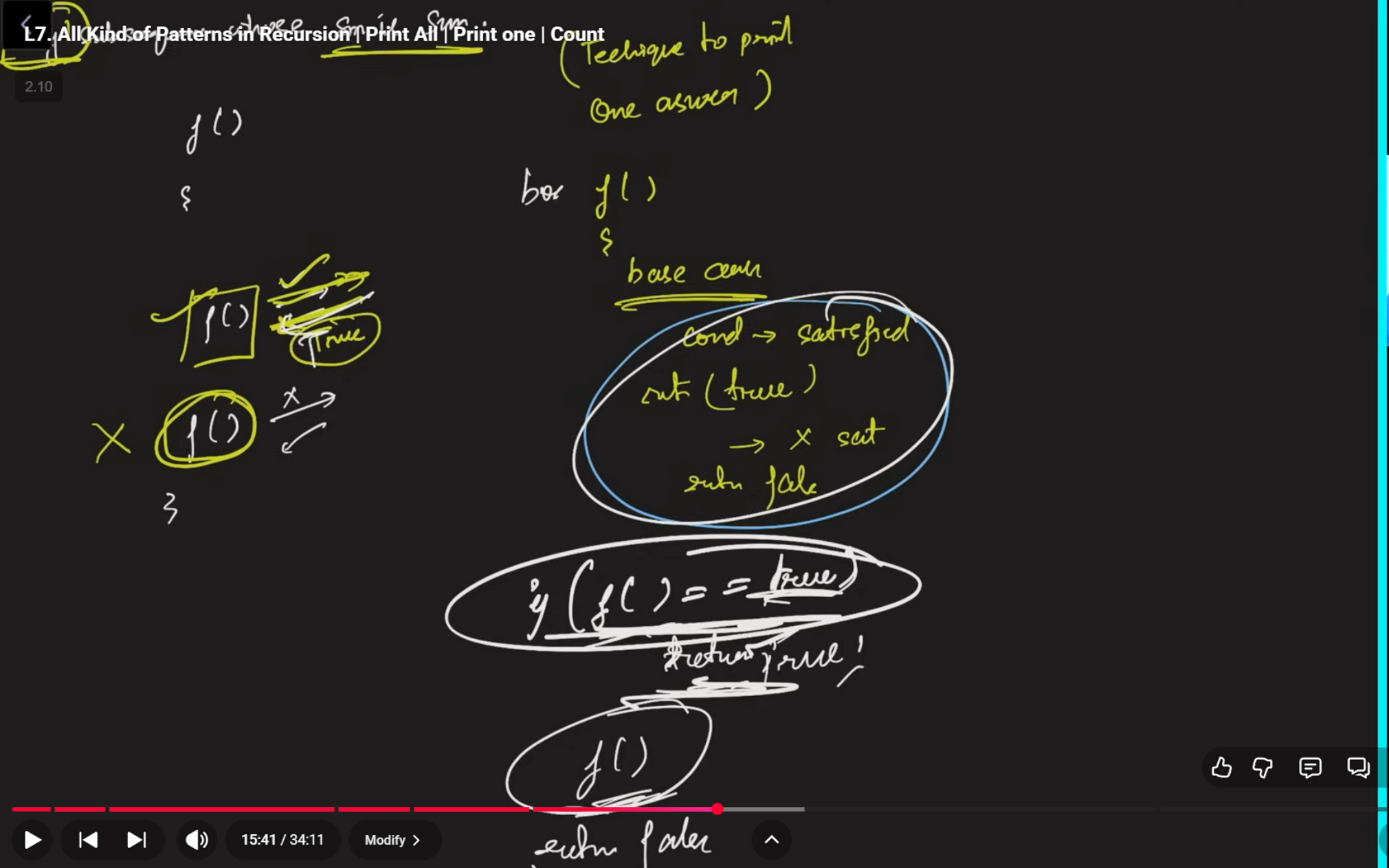
```
5
1 2 3 4 5
```

output.txt

```
1 1
```

```
[Finished in 1.9s]
```

12:34 / 34:11 Modify


(Technique to print One answer)
f()
{
bool f()
{
base case
cond → satisfied
ret (true)
→ X ret return false
if (f() == true)
return true;
f()
return false

code.cpp

```cpp

bool printS(int ind, vector<int> &ds, int s, int sum, int arr[], int n) {
    if(ind == n) {
        // condition satisfied
        if(s == sum) {
            for(auto it : ds) cout << it << " ";
            cout << endl;
            return true;
        }
        // condition not satisfied
        else return false;
    }

    ds.push_back(arr[ind]);
    s += arr[ind];

    if(printS(ind+1, ds, s, sum, arr, n) == true) {
        return true;
    }

    s-= arr[ind];
    ds.pop_back();

    // not pick
    if(printS(ind+1,ds, s, sum, arr, n) == true) return true;

    return false;
}
int main() {
    #ifndef ONLINE_JUDGE
    freopen("input.txt", "r", stdin);
    freopen("output.txt", "w", stdout);
    #endif
    int arr[] = {1, 2, 1};
```

input.txt

```
5
1 2 3 4 5
```

output.txt

```
1 1

```

```
[Finished in 2.2s]
```

2.10

→ Count the subsequences with sum = k

[1, 2, 1] k = 2

(2 subsequences)

2.10

```
int f()
{
    base case

    return 1 → condition satisfies
    return 0 → condition not satisfies

    l = f()
    r = f()
    return l+r;
}
```

2.10

base case

return 1 → condition satisfies

return 0 → condition not satisfies

```
{ l = f()
  r = f()
  return l+r;
}
```

2

```
s = 0
fn(i = 1 → n)
    s++ = f();
```

N times

return s

27. All Kind of Patterns in Recursion | Print All | Print one | Count

```cpp
int printS(int ind, int s, int sum, int arr[], int n) {
    if(ind == n) {
        // condition satisfied
        if(s == sum) return 1;
        // condition not satisfied
        else return 0;
    }


    s += arr[ind];

    int l = printS(ind+1,  s, sum, arr, n) ;

    s-= arr[ind];


    // not pick
    int r = printS(ind+1, s, sum, arr, n);

    return l + r;
}
int main() {
    #ifndef ONLINE_JUDGE
    freopen("input.txt", "r", stdin);
    freopen("output.txt", "w", stdout);
    #endif
    int arr[] = {1, 2, 1};
    int n = 3;
    int sum = 2;
    cout << printS(0, 0, sum, arr, n);

    return 0;
}
```

input.txt

```
1
```

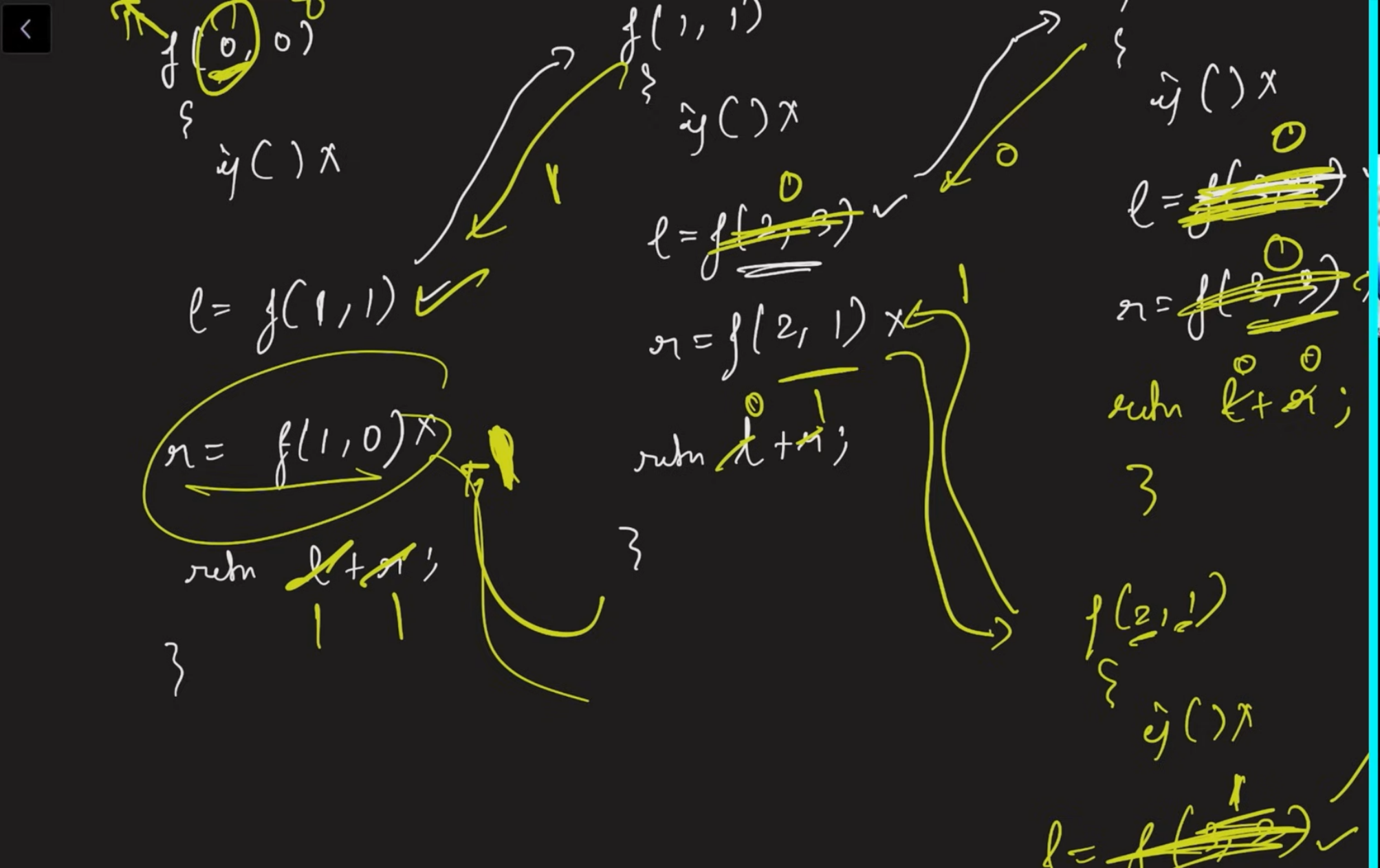
output.txt

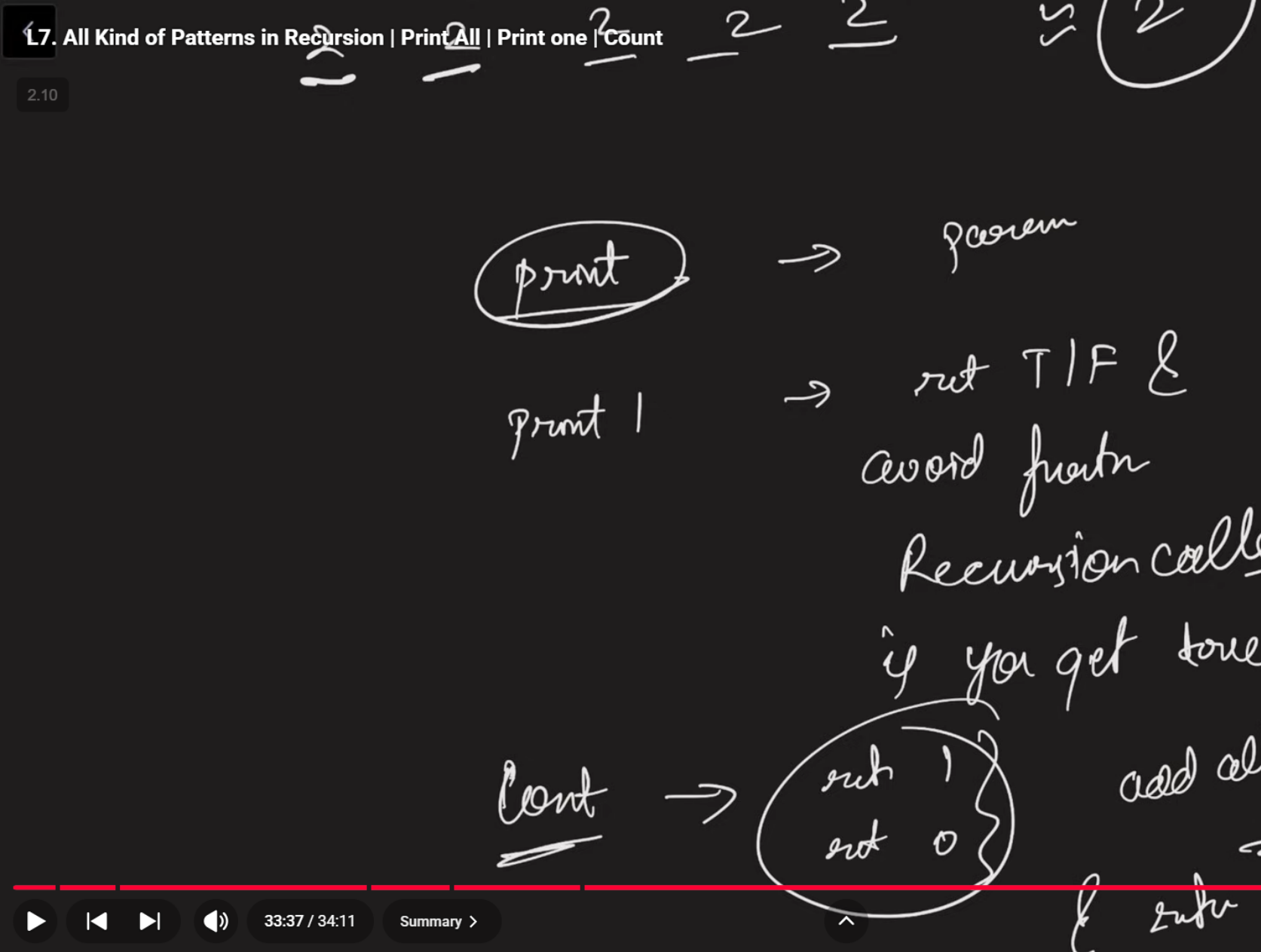
```
2
```

```
[Finished in 0.8s]
```

27:27 / 34:11 Change

6 characters selected Tab Size: 4 C++


f(0, 0)
{
ỳ()x
l = f(1, 1)
f(1, 1)
{
ỳ()x
l = f(2, 1)
r = f(1, 0)x
return l + r;
}
return l + r;
}
f(2, 1)
{
ỳ()x
l = f(1, 0)
r = f(0, 0)x
return l + r;
}
f(2, 1)
{
ỳ()x
l = f(1, 0)
r = f(0, 0)x
return l + r;
}
f(2, 1)
{
ỳ()x
l = f(1, 0)


2.10
print
→
print 1
→
Count
→
if you get true
add all f()
33:37 / 34:11
Summary